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Movie Data Visualization Report

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This analysis explores the central business question: How do budget, rating, and popularity correlate with movie revenue? Using a comprehensive movie dataset, we examine patterns across financial investment, audience reception, and market performance. The goal is to uncover meaningful relationships that can inform production, marketing, and distribution strategies in the film industry.

Dataset Description

The dataset consists of multiple attributes related to movies, including budget, revenue, popularity, vote average (rating), original language, and release date. These variables allow for both univariate and multivariate analysis. Budget and revenue provide financial insight, while rating and popularity reflect audience reception. Release date enables time-series analysis, and original language introduces a cultural dimension.

Design Choices

The visualizations in this analysis were designed with clarity, consistency, and interpretability in mind. Each chart uses: Appropriate scales to ensure accurate representation of data. Consistent color schemes to maintain visual coherence across

charts. Clearly labeled axes to avoid ambiguity. Supporting interpretations below each plot to connect visuals with business insights.

A variety of chart types, bar charts, line graphs, histograms, scatter plots, box plots, heatmaps, and interactive dashboards were selected to highlight different aspects of the data. This diversity ensures that both distributional patterns and relationships between variables are effectively communicated.

Storyline

The analysis begins by exploring popularity across languages, then transitions into budget trends over time, followed by rating distributions. From there, it examines relationships between rating and revenue, seasonal revenue patterns, and yearly revenue trends based on ratings. Finally, it integrates these insights into a comprehensive dashboard to directly answer the business question.

Insights from Visualizations

Popularity Among Languages (Bar Chart)

This visualization shows the popularity among original spoken languages for each movie. It reveals that movies with the original language in Greek (“el”) are the most popular. Meanwhile, Japanese (“ja”) and Indonesian (“id”) films display higher variability, indicating less consistency in popularity. This is important because it highlights how language and cultural origin may influence audience engagement. The key insight is that Greek, Japanese, and Indonesian languages stand out in popularity, though with differing levels of certainty.

Budget Trends Over Time (Line Chart)

This visualization shows the trend of movie budgets across years. It indicates that films released between 1920 and 1940 had notably high budgets. After a decline, budgets increased again from the 1980s through the 2000s, though not reaching earlier peaks. This is important because it identifies historical investment patterns in filmmaking. The key insight is that while early cinema saw significant investment, modern films have experienced a resurgence in budget growth, reflecting industry expansion and technological advancement.

Distribution of Movie Ratings (Histogram)

This visualization shows the distribution of average review ratings. Most movies cluster between ratings of 5 and 7, with fewer films receiving very low ratings. The distribution is slightly negatively skewed, meaning the mean is less than the median, which is less than the mode. This is important because it reveals audience rating behavior. The key insight is that mid-range ratings dominate, suggesting that most films are perceived as average to moderately good.

Rating vs Revenue (Scatter Plot)

This visualization explores the relationship between vote average and revenue. It shows clustering between ratings of 5 and 8, with higher revenues generally associated with higher ratings. This is important because it helps determine whether better-rated movies earn more. The key insight is a mild positive relationship: higher ratings tend to correspond with higher revenue, though the correlation is not strong.

Revenue by Release Month (Box Plot)

This visualization shows the distribution of revenue across release months. Summer and holiday months generate the highest revenues. This is important because it highlights seasonal effects in movie performance. The key insight is that timing plays a critical role films released during peak seasons tend to perform better financially.

Revenue Trends by Rating Over Time (Line Graph)

This visualization shows yearly revenue trends filtered by rating. Movies with ratings between 7 and 8 consistently generate the most revenue. This is important because it connects quality perception with financial performance over time. The key insight is that moderately high-rated films tend to dominate revenue generation.

Rating Influence on Revenue (Joint Plot)

This visualization shows the distribution of revenue based on rating. It confirms that movies with ratings between 6 and 8 generate the most revenue. This is important because it reinforces the relationship between audience satisfaction and earnings. The key insight is that extremely high or low ratings are less associated with top revenue performance.

Popularity Heatmap by Release Date

This visualization maps movie popularity across months and years. It shows that the 2010s had the highest overall popularity, with peaks during summer and holiday months. This is important because it identifies both temporal and seasonal trends. The

key insight is that modern films, supported by stronger advertising and global reach, achieve higher popularity.

Revenue by Rating Category (Bar Graph)

This visualization summarizes revenue across rating categories. Ratings between 5 and 8 generate the most revenue, while very low (below 4) and very high (above 8) ratings generate less. This is important because it challenges the assumption that only top-rated films succeed. The key insight is that broadly appealing, mid-range films often outperform niche or critically extreme films.

Rating and Revenue Over Time (Scatter Plot)

This visualization shows the relationship between rating and revenue across years. Ratings between 6 and 8 consistently generate the highest revenue. This is important because it adds a time dimension to the rating-revenue relationship. The key insight is that mid-range ratings remain the most commercially successful across different periods.

Revenue Distribution by Rating (Interactive Box Plot)

This visualization shows quartiles, medians, and ranges of revenue for each rating. Ratings between 5 and 8 produce the highest revenue distributions, while ratings below 5 and above 9 show limited revenue. This is important because it provides a deeper statistical view. The key insight is that there is little to no strong correlation between extreme ratings and revenue, while mid-range ratings dominate.

Final Business Insight: Budget, Rating, and Revenue Correlation

The interactive dashboard directly addresses the business question by combining a scatter plot of budget vs revenue and a bar chart of rating vs revenue. Budget has a strong positive correlation with revenue: higher-budget films tend to generate higher revenue. Rating has a weak or no strong correlation with revenue: films with ratings between 5 and 8 generate the most revenue, but higher ratings do not guarantee higher earnings.

Conclusion

The analysis reveals that budget is the most significant driver of revenue, while rating and popularity play supporting but less definitive roles. Popularity varies by language and release timing, while ratings tend to cluster in a range that aligns with commercial success. Investing in higher budgets can increase revenue potential. Targeting broad audience appeal may yield better financial outcomes. Strategic release timing especially during summer and holidays can significantly boost performance. In conclusion, while quality and popularity matter, financial investment remains the strongest predictor of box office success.